Week 4 of 6

4 Lending, borrowing & stablecoins

Task	Model a loan to liquidation in a spreadsheet, with a live Pyth price
Blog post	Understanding lending and borrowing in DeFi

Today

1. A bank with no banker
2. The four numbers
3. Liquidation, step by step
4. Where prices come from
5. Stablecoins: the dollars of DeFi

A bank with no banker

Lenders, borrowers, and a program in the middle.

No name, no credit score, no court

A bank lends on

- who you are
- your income and history
- the right to sue you

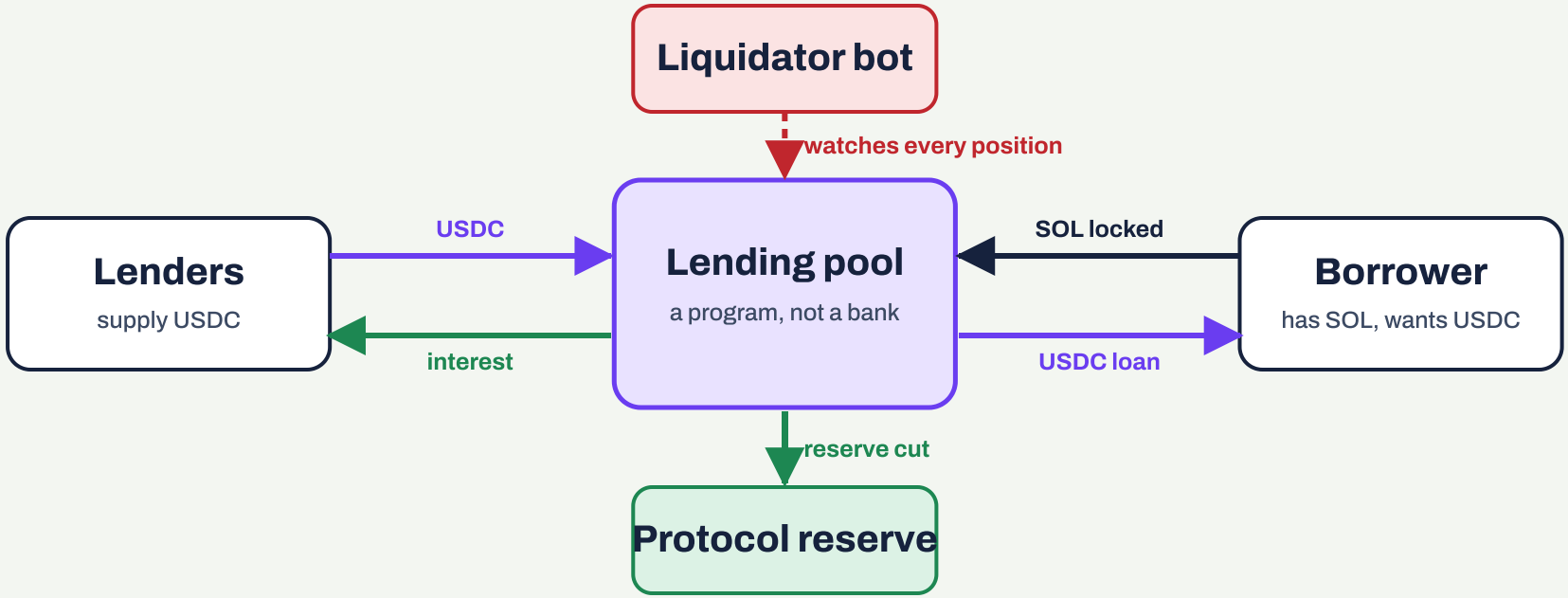
A lending program knows

- an address
- what that address locked up

So every loan is **worth less than its collateral**.

$150 in → borrow at most $105

Who pays whom



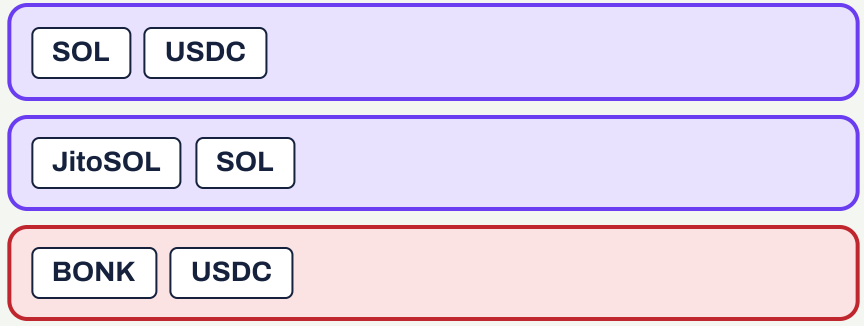
Solana’s lenders

Shared pool



Every asset backs every loan. One bad asset can hurt everyone.

Isolated markets



Each market stands alone. A bad asset only hurts its own market.

Kamino Lend · Jupiter Lend · Save · Project 0

The four numbers

LTV, liquidation threshold, health factor, utilization.

One position, one bar



LTV

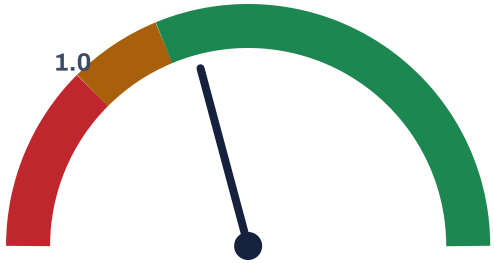
900 ÷ 1,500 = 60%

Health factor

1,500 × 0.80 ÷ 900 = 1.33

Below 1.00 = liquidation

Your position, live



1.33

Healthy

SOL price **$150**

USDC borrowed **$900**

Collateral	**10 SOL = $1,500**
LTV	**60.0%**
Liquidation price	**$112.50**
Can still borrow	**$150**

```
HF = collateral × 0.80 ÷ debt
```

Liquidation

What happens at a health factor below 1.

The walkthrough



Collateral

5.705 SOL = $627.55

Debt

$450

LTV

71.7%

Health

1.12

Liquidator bot

Repays **450 USDC** of debt (50% close factor)

Takes $450 + 5% bonus = **$472.50 of SOL** = 4.295 SOL

Profit: **$22.50**, before fees

Borrower after

Lost 4.295 SOL. Still owes $450. Health back above 1, at 1.12.

The 5% bonus is the penalty they paid.

Flash loans: a million, for one transaction

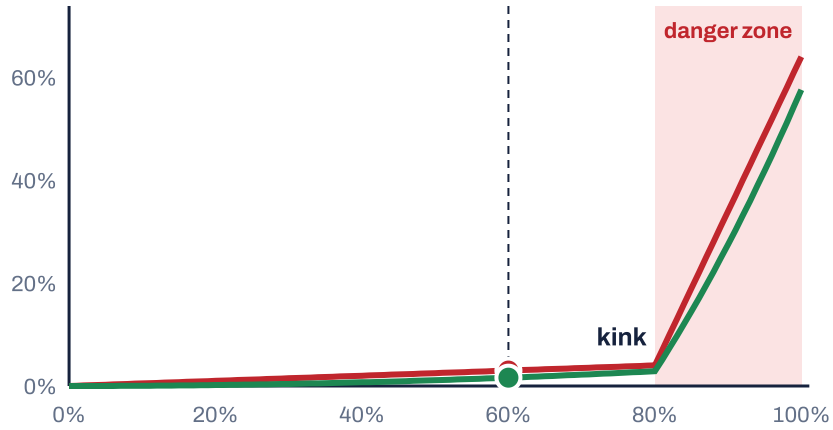
One transaction, all or nothing

1. Borrow **1,000,000 USDC**
   no collateral
2. Use it
   repay someone's debt in a liquidation, or buy low here, sell high there
3. Can't repay
   the program refuses to finish

Reverted

As if steps 1 and 2 never happened. Only the network fee is spent.

What borrowing costs



Utilization **60%**

`utilization = borrowed ÷ supplied`

Borrowers pay **3.0%**

Lenders earn **1.6%**

Plenty of cash in the pool. Rates stay low.

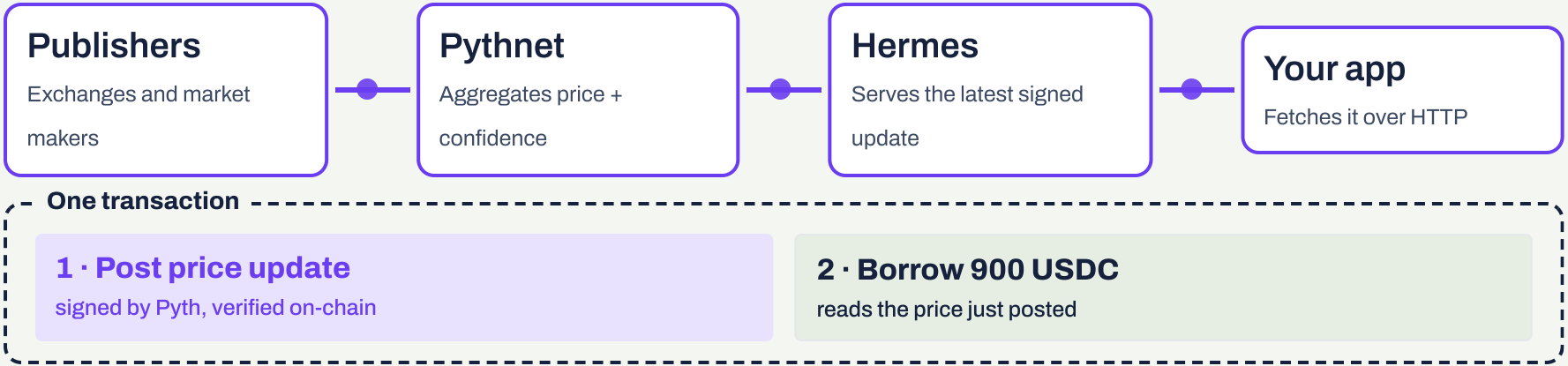
10 SOL collateral, 900 USDC debt, liquidation threshold 80%. At what SOL price can you be liquidated?

- **A** $90.00
- **B** $112.50
- **C** $128.57
- **D** $150.00

Where prices come from

A lender is only as safe as its price feed.

How Pyth delivers a price



Program checks: **signature valid** · **feed is SOL/USD** · **not too old**, then prices your collateral.

`price: 15234000000, expo: -8` → $15234000000 \times 10^{-8} = \$152.34$

See the live price

Users

```
insights.pyth.network
```

Search **SOL/USD**. Read the price, the confidence band and when it last updated.

Builders

```
npm run w4:price
```

Reads Pyth's SOL/USD account on Solana: `price × 10^expo`, after checking who owns the account.

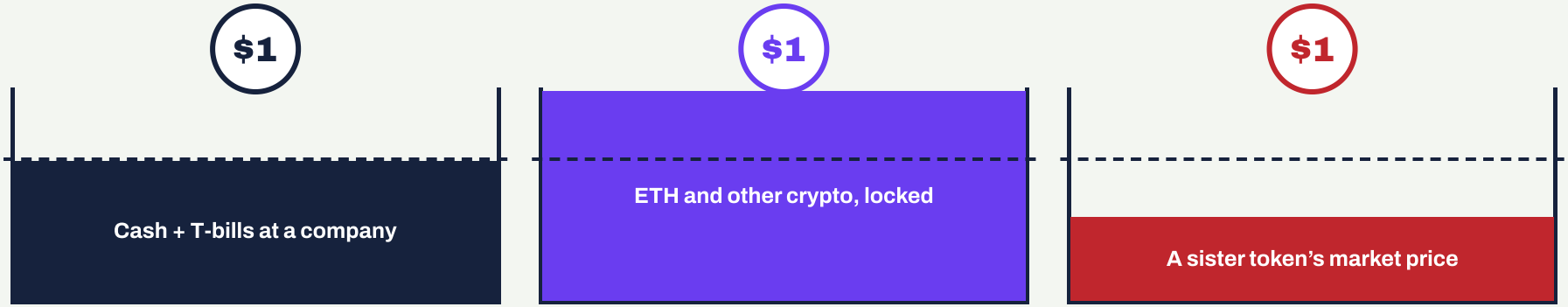
> **Hermes now needs a key**
> Since 26 Aug 2026 the public Hermes URL returns 401. Apps use a key; you don't need one for class.

Write down **the price and the time you looked**. That's your spreadsheet's starting row.

Stablecoins

The dollars of DeFi.

Three ways to hold $1



Fiat-backed

USDC · USDT · PYUSD · USDG

Peg: Redeem 1 coin for $1 with the issuer

Breaks when: Issuer, bank or reserve problem. Coins can be frozen.

Crypto-backed

DAI, now Sky (history)

Peg: Over-collateralized vaults, liquidated if they slip

Breaks when: Collateral crashes faster than liquidations run.

Algorithmic

UST (Terra, collapsed May 2022)

Peg: Burn $1 of the sister token to mint 1 coin

Breaks when: Confidence breaks, both tokens spiral to zero.

What holds the peg



Market price **$0.970**

Issuer is redeeming

Buy at $0.970, redeem for $1.00

Profit 3.0¢ per coin. Buying lifts the price; redeeming burns supply.

Why it matters here

Payments

Settles in under a second for a fraction of a cent, any day of the week.

Savings

A dollar balance without a dollar bank account.

Remittance

Wallet to wallet across borders, no correspondent bank.

The catch

Off-ramps, local rules, issuer freezes, and scams.

USDC trades at $0.96 and Circle is redeeming normally. What pulls it back to $1?

- **A** Circle prints more USDC
- **B** Traders buy at $0.96 and redeem for $1
- **C** A Solana validator vote
- **D** Nothing, it stays there

Task: take a loan to liquidation

Users

Simulate it (recommended)

1. Get the live SOL price (insights.pyth.network)
2. Spreadsheet: deposit, borrow, then drop the price in steps
3. Each row: LTV, health factor
4. Mark the liquidation price and the liquidator's profit

Builders

Mainnet fork

1. `surfpool start`: real Kamino and Jupiter Lend accounts, locally
2. Fund a wallet with cheatcodes
3. Deposit and borrow through the SDK
4. Move time and price, watch health

> **Optional, real money**
> Already hold USDC? Supply $5 to a lender and watch interest accrue. Supply only. Never borrow.

Blog post

"Understanding lending and borrowing in DeFi"

Include your liquidation walkthrough, and one paragraph on which stablecoin you'd trust, and why.

Teacher prep

For you, before class. Skip this live.

You, before class

1. **Learn the four numbers.** LTV, liquidation threshold, health factor, utilization. Then the kinked rate curve.
2. **Walk one liquidation by hand.** 10 SOL at $150, borrow 900 (60%). SOL to $110: LTV ≈ 82%.
3. **Look up SOL's real liquidation threshold** on Kamino or Jupiter Lend. Who gets liquidated, and how does the liquidator profit?
4. **Read Pyth's Solana pull-integration guide.** Week 5 depends on it.
5. **Know the stablecoin types,** and which have real depth on Solana.

Verify before class

Any devnet lending market

No lender has a public devnet UI. Builders: create a Kamino devnet market with klend-sdk, or fork mainnet with Surfpool.

marginfi is now Project 0

Same team and program, renamed Sep 2025. Old app shut in mid-2026; use app.0.xyz.

Numbers with dates

Stablecoin supply ($17.3B, 25 Sep 2026) and TVL move weekly. Re-check on DefiLlama the day you teach.

Builders' fork

`surfpool start` · port 8899 · cheatcodes for balances and time